

# Lecture 10

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## *Theory of Square Linear System III: Existence and Expression of Inverse*

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# Part 0 Review and Outline

6 mins

# Review: **No-Row-Exchange Case**: LU Decomposition

Coefficient matrix **A**; square matrix.

(S1) Gauss-Jordan elimination (forward part; assume no row exchange)

$$A \rightarrow A_1 \rightarrow A_2 \dots \rightarrow U$$

U is upper triangular.

(S2) Express GE as matrix multiplication:

$$E_k E_{k-1} \dots E_1 A = U.$$

Express matrix **A** as matrix product:

$$A = (E_k E_{k-1} \dots E_1)^{-1} U.$$

Remark: This expression holds for any square matrix A.

“No row exchange” is used in the next step.

(S3) Denote  $L = (E_k E_{k-1} \dots E_1)^{-1}$ ; it is lower triangular.

since only involves Type-III (eliminating down) and Type-II elementary matrices.

Then

(S4) 
$$A = LU.$$

# Review: 3 by 3 Example: Compute LU

**Step 1 Gauss-Jordan elimination (only the forward elimination part)**

$$\begin{bmatrix} 2 & 4 & 2 \\ 1 & 5 & 2 \\ 4 & -1 & 9 \end{bmatrix} \xrightarrow[\substack{R_2 \rightarrow -\frac{1}{2}R_1 + R_2 \\ R_3 \rightarrow -2R_1 + R_3}]{\substack{R_2 \rightarrow -\frac{1}{2}R_1 + R_2 \\ R_3 \rightarrow -2R_1 + R_3}} \begin{bmatrix} 2 & 4 & 2 \\ 0 & \boxed{3} & 1 \\ 0 & -9 & 5 \end{bmatrix} \xrightarrow{R_3 \rightarrow -(-3)R_2 + R_3} \begin{bmatrix} 2 & 4 & 2 \\ 0 & \boxed{3} & 1 \\ 0 & 0 & 8 \end{bmatrix} \triangleq U.$$

Check: No row exchange in the process; continue.

**Step 2: Record elementary matrices.**

$$E_1, E_2, E_3 = \begin{bmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$$

**Step 3: Compute L by (10.3).**

$$\begin{aligned} L &= E_1^{-1} E_2^{-1} E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}^{-1} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 2 & -3 & 1 \end{bmatrix} \end{aligned}$$

**Step 4: Conclusion.**

A has an LU decomposition  $A = LU$ , where  $L = \text{blabla}$ ,  $U = \text{blabla}$ .

# Today's Lecture: Key Questions

Consider a square linear system  $A\mathbf{x} = \mathbf{b}$

**Theorem** If  $A$  is invertible, then the linear system has a unique solution  $\mathbf{x} = A^{-1}\mathbf{b}$ .

**Question 1:** When is  $A$  invertible?

**Question 2:** How to Express/Compute  $A^{-1}$ ?  
(if exists)

Will provide an answer today.

# Today's Lecture: Outline

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Today ... **Existence** and **expression**/computation of inverse.

1. Existence of Inverse
2. Expressions and Computation of inverse

Strang's book: Sec 2.5, 2.6

After this lecture, you should be able to

1. Tell when a matrix is invertible based on pivots and equation solving
2. Express the inverse of an invertible matrix with the aid of elimination
3. Compute the inverse of a small matrix
4. Prove a few results related to the test of inverse

Remark:

We'll show the proofs in detail, so as to help you understand how to prove things.

# Comments on Course Contents Arrangement

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## *Theme of Lec 3-15: Solving Linear Systems*

(along with Theory of Matrix & Linear Space)

### Course arrangement:

In Lec 3,4,7, we learned GJ-E for solving square linear systems.

In Lec 8-10, we mainly learn two things:

1) Matrix inverse. (Lec 8 & 10)

This can be used for representing solution of good square linear systems.

2) LU decomposition. (Lec 9)

This is a generic result on square matrices.

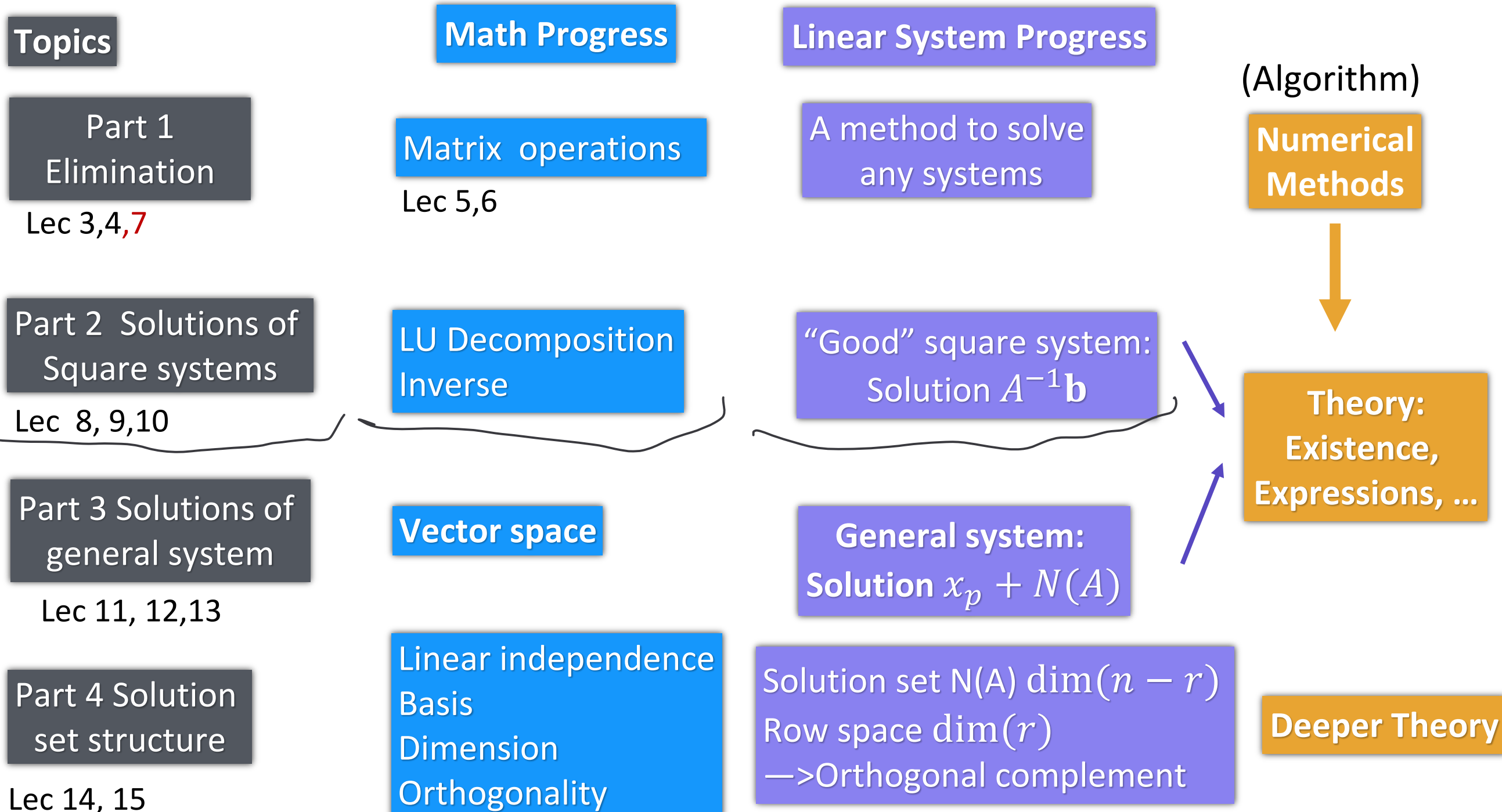
We will use tools on matrix multiplication from Lec 5-6.

Remark: This slide appeared in Lec 8.

# Roadmap: *Topics Before Midterm*

Lec 1,2: Vector operations.

*Theme of Lec 3-15: Solving Linear Systems* (along with Theory of Matrix & Linear Space)



# Part I Inverse of A: Existence

Strang's book v5, Sec 2.5, Segment "Singular versus Invertible"

## Part I.1 Test by Pivots

25 mins

# When is $A$ Invertible?

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**Question 1:** When is  $A$  invertible?

# When is $A$ Invertible?

## Question 1: When is $A$ invertible?

Suppose  $A$  is an  $n \times n$  matrix.

We use Gauss-Jordan elimination to solve  $Ax = b$ , where  $b$  is an arbitrary vector.

**Proposition 1:** If  $n$  pivots exist in Gauss-Jordan elimination, then  $A$  is invertible.

**Proposition 2:** If  $A$  is invertible, then there are  $n$  pivots in Gauss-Jordan elimination.

**Theorem 10.1:**  $A$  is invertible iff  $A$  has  $n$  pivots (assuming  $A$  is  $n$  by  $n$  matrix).

See also Sec 2.5 of Strang's book;  
2nd bullet in the beginning of Sec 2.5.

# How To Come Up with the Proof

**Goal:**  $A$  is invertible iff  $n$  pivots exist in Gauss-Jordan elimination

*Tree of thought*

Tip: Start with one concept, and try to extend.

Inverse



?

GJ-E

?



Choose which to extend?

# How To Come Up with the Proof

**Goal:**  $A$  is invertible iff  $n$  pivots exist in Gauss-Jordan elimination

Tip: Start with one concept, and try to extend.

Inverse

def:  $AB=BA=I$ ,  
related to product

GJ-E

A bunch of  
row operations

A bunch of matrix  
multiplications

?

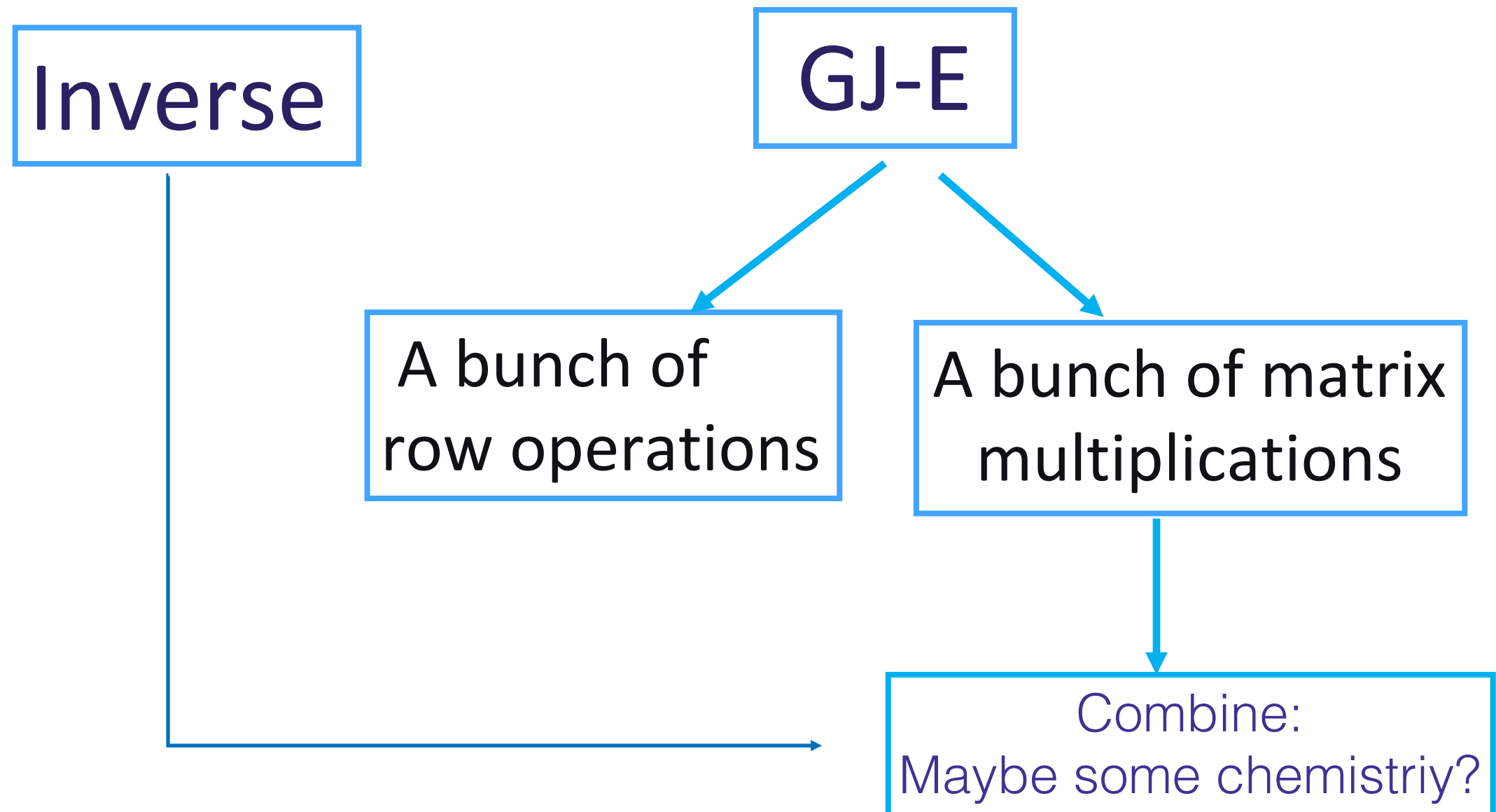
?

Choose which to extend?

# How To Come Up with the Proof

**Goal:**  $A$  is invertible iff  $n$  pivots exist in Gauss-Jordan elimination

Tip: Start with one concept, and try to extend.



# Proof of Prop. 1: Step 1

## A as Product of Row Elementary Matrices

If there are  $n$  pivots (possibly with row exchange):

**Step 1: GJ-E can be represented as matrix multiplication.**

$$\underbrace{A \xrightarrow{\bar{E}_1} A_1 \xrightarrow{\bar{E}_2} A_2 \dots \rightarrow U}_{\text{GJ-E process}} \rightarrow \dots B_1 \rightarrow \dots \xrightarrow{\bar{E}_p} I_n \quad (n \text{ pivots})$$

Handwritten notes:

$$A_1 = \bar{E}_1 A$$
$$A_2 = \bar{E}_2 A_1 = \bar{E}_2 \bar{E}_1 A$$
$$\vdots$$
$$U = \begin{bmatrix} 1 & * & \dots & * \\ 0 & 1 & \dots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix} = I_n$$

Corresponding elementary matrices:  $E_p, E_{p-1}, \dots, E_1$

Then the matrix representation of the whole GE process is

$$\bar{E}_p \bar{E}_{p-1} \dots \bar{E}_2 \bar{E}_1 A = I_n$$

$$\underline{Q}A = I_n, \text{ where } Q = \bar{E}_p \bar{E}_{p-1} \dots \bar{E}_1$$

**Step 2:**

$$QA = I_n \stackrel{?}{\Rightarrow} A^{-1} \exists.$$

[just one-side equation;  
need extra argument.  
Next slide]

# Proof of Proposition 1: Step 2

## Left Inverse is Invertible $\implies$ Itself Invertible

$$QA = I_n \quad (1) \quad \stackrel{?}{\implies} \quad A^{-1} \exists.$$

Recall:  $M \overset{M^{-1}}{\underbrace{A}_{\downarrow \text{unknown}}} = \overset{M^{-1}}{U} \implies A = M^{-1}U$  [Lec 9]

Now:  $\overset{(Q^{-1})}{Q} \overset{(Q^{-1})}{\underbrace{A}_{\text{unknown}}} = \overset{(Q^{-1})}{I} \implies A = \boxed{Q^{-1}} \quad (2) \text{ if } \underline{Q} \text{ is } \boxed{\text{invertible}}.$

# Proof of Proposition 1: Step 2

## Left Inverse is Invertible $\implies$ Itself Invertible

$$QA = I_n \quad (1) \quad \stackrel{?}{\implies} \quad A^{-1} \exists.$$

Recall:  $MA = U \implies A = M^{-1}U$  [Lec 9]  $M = \tilde{E}_k \dots \tilde{E}_1$   
 $M^{-1} \exists : \tilde{E}_i^{-1} \exists \text{ \& (prod of inv) }^{-1} \exists.$

Now:  $QA = I \implies A = Q^{-1}$  (2) if  $Q$  is invertible.

Here  $Q = \overset{\text{product of } E_i\text{'s}}{E_p E_{p-1} \dots E_1}$ , where  $E_i$ 's are elementary matrices.

$Q$  is invertible because elementary matrices are invertible ( $\implies \tilde{E}_i^{-1} \exists$ ),  
and product of invertible matrices is invertible. [Lec 8 final property]

(2)  $\implies$   $A = Q^{-1}$   $\implies$   $AQ = Q^{-1}Q = I_n$   
 $QA = I_n$   $\implies$   $A^{-1}$  exists and  $A^{-1} = Q$   
 (by def of inv).

# Left Inverse and Right Inverse [extended Reading]

## Definition (Left Inverse)

Let  $A$  and  $B$  be matrices.

If  $BA = I$ , where  $I$  is the identity matrix, then  $B$  is called the **left inverse** of  $A$ .

## Definition (Right Inverse)

Let  $A$  and  $B$  be matrices.

If  $AB = I$ , where  $I$  is the identity matrix, then  $B$  is called the **right inverse** of  $A$ .

## Recall:

### Definition of Inverse

If  $B$  is both the left inverse and the right inverse of  $A$ , that is,  $BA = I$  and  $AB = I$ , then  $B$  is the inverse of  $A$ , denoted as  $A^{-1} = B$ .

Note:  $QA = I_n$  (1) itself only implies  $Q$  is a **left inverse** of  $A$ ;   
so need extra argument.

## Left inverse v.s. Inverse [extended Reading]

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**Conjecture 1:** If square matrices  $M, A$  satisfy  $MA = I_n$ ,  
then  $A^{-1}$  exists and equals  $M$ .

In fact, Conjecture 1 holds; but we'll not prove it today.  
so let's still call it Conjecture 1.

Don't directly use Conjecture 1 in homework/exam  
unless you are told so.

The proof of Conjecture 1 is harder and utilize  
many ingredients used in proving Lemma 1.

# Summary: Proof of Proposition 1

**Proposition 1:** If  $n$  pivots exist in Gauss-Jordan elimination, then  $A$  is invertible.

Summary of Proof:

$n$  pivots exist

matrix representation of GJ-E

$$E_p E_{p-1} \dots E_1 A = I_n$$

Inverse of  $E_p E_{p-1} \dots E_1$  exist

Elementary matrices are invertible

product of invertible matrices is invertible

$$A^{-1} \text{ exists and } A^{-1} = E_p E_{p-1} \dots E_1$$

## Proof of Prop. 2: 【Skip】

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**Prop. 2:** If  $A$  is invertible, then there are  $n$  pivots in Gauss-Jordan elimination.

**Proof:** Skipped.

(See Strang's book 5th edition page 88,  
after "Reasoning in reverse")

Need to prove by contradiction (反证法).

# Logic Class: Contrapositive Statement (逆否命题)

Statement S: If P, then Q. [ $P \Rightarrow Q$ ]

Contrapositive statement T: If not Q, then not P. [ $\neg Q \Rightarrow \neg P$ ]

**Important fact:** S and T are equivalent.

If S is true, then T is true.

If T is true, then S is true.

# Logic Class: Contrapositive Statement (逆否命题)

**Statement S:** If P, then Q.  $[P \Rightarrow Q]$

**Contrapositive statement T:** If not Q, then not P.  $[\neg Q \Rightarrow \neg P]$

**Important fact:** S and T are equivalent.

If S is true, then T is true.

If T is true, then S is true.

**Example 1: [both True]**

**Original:** If it is raining (P), then the ground is wet (Q).

**Contrapositive:** If the ground is **not** wet ( $\neg Q$ ), then it is **not** raining ( $\neg P$ ).

**Example 2: [both false]**

**Original:** If it is a bird (P), then it can fly (Q).“ [Counter-example: Penguin]

**Contrapositive:** "If it can't fly ( $\neg Q$ ), then it's not a bird ( $\neg P$ )."

For more: i) Check the PDF “1\_Reading\_basic\_logic\_PreparingForLinearAlgebra” sent in Week 2;

ii) Check blog “Converse, Inverse, and Contrapositive” <https://www.themathdoctors.org/converse-inverse-and-contrapositive/>

## Exercise: Judge T or F

J1 If an  $n \times n$  matrix  $A$  is invertible, then there are  $n$  pivots in Gauss-Jordan elimination for solving  $Ax = 0$ . Prop. 2.

T

J2 If an  $n \times n$  matrix  $A$  is not invertible, then there are less than  $n$  pivots in Gauss-Jordan elimination for solving  $Ax = 0$ .

T

Contrapositive of Prop. 1:  $n \text{ pivots} \Rightarrow A^{-1} \exists$   
 $A^{-1} \nexists \Rightarrow < n \text{ pivots}$

J3 Suppose  $A$  is an  $n \times n$  matrix  $A$ . If there are less than  $n$  pivots in Gauss-Jordan elimination for solving  $Ax = 0$ , then  $A$  is invertible.

not invertible

F

日期: /

$$\boxed{A^{-1} \exists} \iff \boxed{n \text{ pivots}}$$

$$\boxed{A^{-1} \nexists} \iff \boxed{< n \text{ pivots}}$$

# Part I Inverse of $A$ : Existence

Strang's book v5, Sec 2.5, Segment "Singular versus Invertible"

## Part I.2 Test by Equation

15 mins

# When is $A$ Invertible?

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**Question 1:** When is  $A$  invertible? 2<sup>nd</sup> answer

**Theorem 10.2:**  $A$  is invertible iff  
 $Ax = 0$  has a unique solution  $x = 0$ .

See also Sec 2.5 of Strang's book;  
4th bullet in the beginning of Sec 2.5.

# When is $A$ Invertible?

**Question 1:** When is  $A$  invertible? 2<sup>nd</sup> answer

**Theorem 10.2:**  $A$  is invertible iff  
 $Ax = 0$  has a unique solution  $x = 0$ .

See also Sec 2.5 of Strang's book;  
4th bullet in the beginning of Sec 2.5.

**Recall:** For any “iff” (if and only if) kind of result,  
need to prove “if part” and “only if” part.

**Pattern:** “ $P$  iff  $Q$ ” shall split into:

- (1) “if part”: “ $P$  if  $Q$ ”, or equivalently “If  $Q$ , then  $P$ ”.
  - (2) “only if part”: “ $P$  only if  $Q$ ”, or equivalently “if  $P$ , then  $Q$ ”.
- [this require some thinking]

# When is A Invertible?

Theorem 10.2:  $A$  is invertible iff  $Ax=0$  has a unique solution  $x=0$ .

$\downarrow$   $P$

$\downarrow$   $Q$

**Proof of “Only if part”:**

$P$  only if  $Q \Leftrightarrow$  If  $P$ , then  $Q$

**Goal:** Want to prove

If  $A$  is invertible, then  $Ax=0$  has a unique solution  $x=0$

This is proved in Lec 8.

## Step 0 of “If Part”: Contrapositive Statement

Theorem 10.2:  $A$  is invertible iff  $Ax=0$  has a unique solution  $x=0$ .

### Proof:

(1) “If” part:

**Goal:** Want to prove

If  $Ax=0$  has unique solution  $x=0$ , then  $A^{-1} \exists$ .

Think: from the condition above, what can you derive?

Contrapositive statement [逆否命题]:

If  $A^{-1} \nexists$ , then  $Ax=0$  does NOT have unique solution  $x=0$ .

Next, prove this contrapositive statement.

# Proof of Contrapositive Statement of "If Part"

Theorem 10.2:  $A$  is invertible iff  $Ax=0$  has a unique solution  $x=0$ .

$\tau_{k \times m}^{12,1}$   
 $(\Rightarrow)$  J2 If an  $n \times n$  matrix  $A$  is **not invertible**, then there are **less than  $n$  pivots** in Gauss-Jordan elimination for solving  $Ax = 0$ .  $\top$

## Proof of "If" part:

Want to prove the

contrapositive statement:

If  $A^{-1} \nexists$ , then  $Ax = 0$  does **not have a unique solution**. ← goal.

## Proof strategy: extending bridge.

- Try to get a new result from the condition.
- Continue: try to get another result.

$$A^{-1} \nexists$$

by Theorem 10.1

sth we learned very recently

$< n$  pivots when solving  $Ax=0$

by the relation of # of pivots  
and # of solutions

something we learn in Lec 7

$Ax=0$  has no or  $\infty$ -many solutions.

## Exercise: Judge T or F

Suppose  $A$  is an  $n \times n$  matrix in the following statements.

J1 If  $Ax = 0$  has infinitely many solutions, then  $A$  is not invertible.

T

J2 If  $A$  is invertible, then  $Ax = 0$  has exactly one solution.

T

J3 If  $A$  is not invertible, then  $Ax = 0$  has more than one solutions.

T

original:  $A^{-1} \nexists \Rightarrow$  no or  $\infty$  many solutions

but  $x=0$  is always a solution of  $Ax=0$ , so  
"no solution" is impossible for  $Ax=0$ .

# Part II Computing Inverse

Strang's book v5, Section “Calculating  $A^{-1}$  by ...”

25 mins

# Outline of This Part

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Outline of this Part:

- Inverse computation I: Naive way.
- Inverse computation II:  
Smarter way by row operations.

# Recall: A as Product of Row Elementary Matrices

Gaussian-Jordan elimination (GJ-E): case of  $n$  pivots

$$A \rightarrow A_1 \rightarrow A_2 \dots \rightarrow U \rightarrow \dots B_1 \rightarrow \dots \dots \rightarrow I_n$$
$$U = \begin{bmatrix} 1 & * & \dots & * \\ 0 & 1 & \dots & * \\ \vdots & \vdots & \ddots & * \\ 0 & 0 & \dots & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \dots & 1 \end{bmatrix} = I_n.$$

Suppose the corresponding elementary matrices are

$$E_p, E_{p-1}, \dots, E_1.$$

Then the matrix representation of the whole GE process is

$$E_p E_{p-1} \dots E_1 A = I_n.$$

$$A^{-1} = E_p \dots E_2 E_1 \quad (10.1)$$

**Question 2a:** How to Express  $A^{-1}$ ?

**Answer:** (10.1)

**Claim:** A matrix is invertible iff it can be written as the product of elementary matrices.

# Recall: Quadratic Equations

During middle school, when you learn quadratic equations, what do you learn?

$$(1) x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Expression (of roots)  
→ algorithm

(2) To solve  $x^2 + 4x + 1 = 0$ ,  
write it as  $(x + 2)^2 = 3$ ,  
Then get  $x + 2 = \sqrt{3}$  or  $-\sqrt{3}$ .

Algorithm or method  
to compute roots.  
(no closed-form expression)

I know you know these. But...

**WHAT** are them?

# Recall: Quadratic Equations

During middle school, when you learn quadratic equations, what do you learn?

$$(1) x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Expression (of roots)

→ can become an algorithm to compute roots

$$(2) \text{ To solve } x^2 + 4x + 1 = 0,$$
$$\text{write it as } (x + 2)^2 = 3,$$
$$\text{Then get } x + 2 = \sqrt{3} \text{ or } -\sqrt{3}.$$

Algorithm (to compute roots)

I know you know these. But...

**WHAT** are them?

# Computing Inverse

## Question 2b: How to Compute $A^{-1}$ ? (if exists)

### Algorithm 1 (compute $A^{-1}$ )

#### Step 1: Forward elimination.

Run forward elimination, till get upper triangular matrix  $U$ .

IF  $U$  contains zero diagonal entry:  $\rightarrow < n$  pivots  
STOP and report: No inverse.

ELSE Go to Step 2.

#### Step 2: Backward substitution.

Run backward substitution, till get identity matrix  $I_n$ .

#### Step 3 Record elementary matrices.

Record elementary matrices  $E_1, \dots, E_k$  in Step 1.

Record elementary matrices  $E_{k+1}, E_{k+2}, \dots, E_p$  in Step 2.

#### Step 4: Compute inverse.

Compute  $A^{-1} = E_p \dots E_2 E_1$ . (10.1)

# Example: Computing Inverse by Naïve Algorithm 1

## Step 1 Gauss-Jordan elimination

Forward elimination (get  $U$ )

$$\begin{bmatrix} 1 & 6 & 0 \\ 1 & 7 & 1 \\ 0 & 1 & 3 \end{bmatrix} \xrightarrow{R_2 \rightarrow (-1)R_1 + R_2} \begin{bmatrix} 1 & 6 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 3 \end{bmatrix} \xrightarrow{R_3 \rightarrow (-1)R_2 + R_3} U = \begin{bmatrix} 1 & 6 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

Backward substitution (get  $I_3$ )

$$U \xrightarrow{R_3 \rightarrow \frac{1}{2}R_3} \begin{bmatrix} 1 & 6 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_2 \rightarrow (-1)R_3 + R_2} \begin{bmatrix} 1 & 6 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \rightarrow (-6)R_2 + R_1} I_3$$

## Step 2: Record elementary matrices.

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}, E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}, E_4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}, E_5 = \begin{bmatrix} 1 & -6 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

## Step 3: Compute $A^{-1}$ by (10.1).

$$A^{-1} = \begin{bmatrix} 10 & -9 & 3 \\ -3/2 & 3/2 & -1/2 \\ 1/2 & -1/2 & 1/2 \end{bmatrix}.$$

Trick 1: Will not use matrix multiplication.  
just use row operations to compute  
The product.

## Motivating Algorithm 2

Trick: Use row operations to compute  $E_p \dots E_2 E_1$ .

This trick is similar to the trick of computing  $E_1^{-1} E_2^{-1} E_3^{-1}$  in LU decomposition.

**Difference:** Here, we compute products of  $E_i$ , not  $E_i^{-1}$ .

**This means:** these row operations are the same as those in GJ-E!

# Motivating Algorithm 2

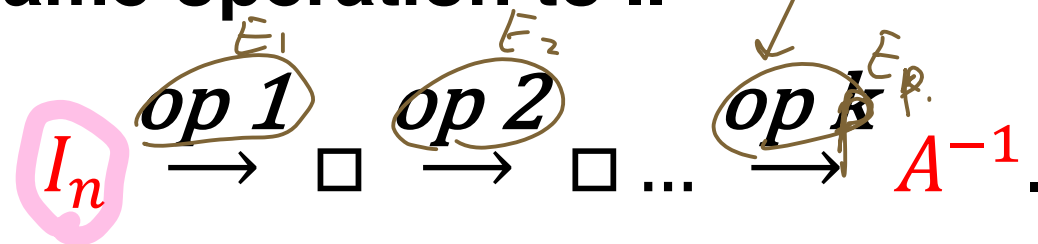
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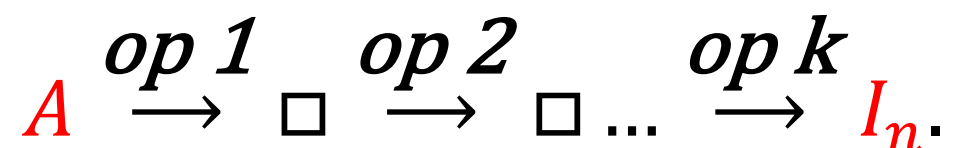
**Difference:** Here, we compute products of  $E_i$ , not  $E_i^{-1}$ .

**This means:** these row operations are **the same as those in GJ-E!**

Module 2: Apply same operation to I.



Module 1: GE.



Same  
Operations!

# Motivating Algorithm 2: Utilize “Same Operations”

Module 2: Apply same operation to I.

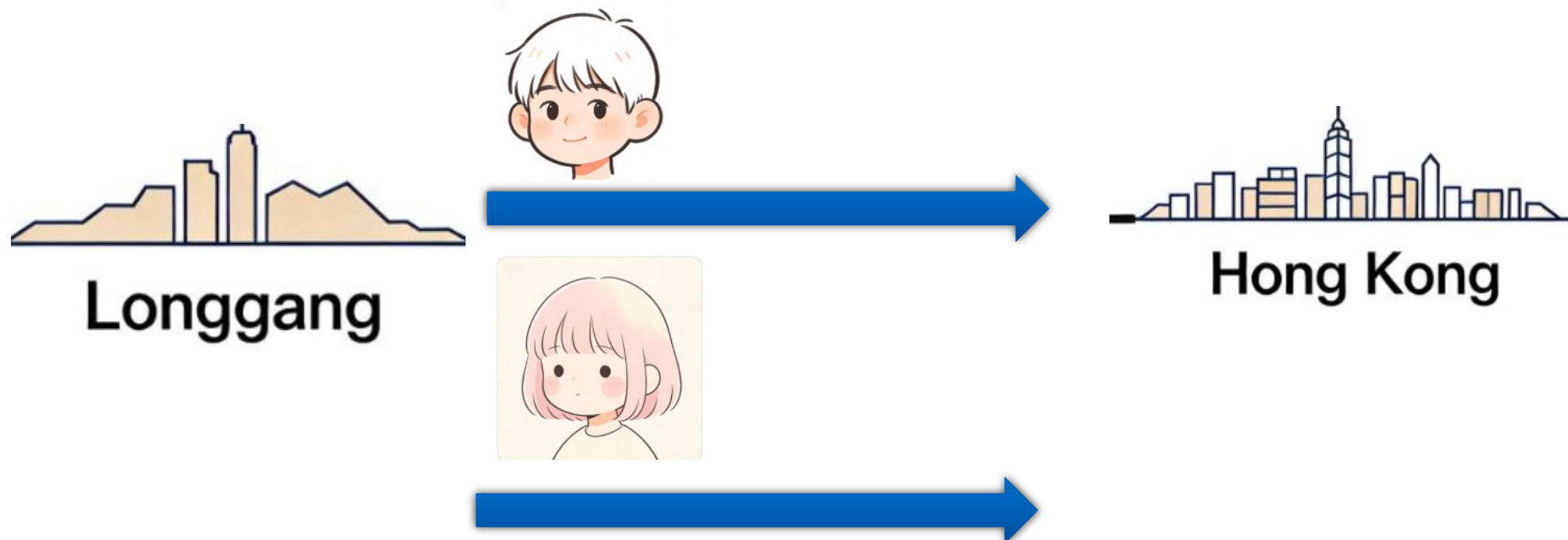
$$I_n \xrightarrow{op\ 1} \square \xrightarrow{op\ 2} \square \dots \xrightarrow{op\ k} A^{-1}.$$

Module 1: GE.

$$A \xrightarrow{op\ 1} \square \xrightarrow{op\ 2} \square \dots \xrightarrow{op\ k} I_n.$$

Same  
Operations!

Both Adam and Cherry want to go to HK.



What to do? Take a taxi together.

# Algorithm 2: Apply GJ-E to A and I Together

Module 1: GE.

$$A \xrightarrow{op\ 1} \square \xrightarrow{op\ 2} \square \dots \xrightarrow{op\ k} I_n.$$

Module 2: Apply same operation to I.

$$I_n \xrightarrow{op\ 1} \square \xrightarrow{op\ 2} \square \dots \xrightarrow{op\ k} A^{-1}.$$

Same  
Operations!

Algorithm 2: Apply GJ-E to  $[A, I_n]$ , together

$$[A, I_n] \xrightarrow{op\ 1} \square \xrightarrow{op\ 2} \square \dots \xrightarrow{op\ k} [I_n, A^{-1}].$$

Another explanation:

i) To solve  $Ax = b$ , from  $[A | b] \xrightarrow{\text{GJ-E}} [I | A^{-1}b]$

ii) To “solve”  $AX = I$ , from  $[A | I] \xrightarrow{\text{GJ-E}} [I | A^{-1}I]$

multiply  $A^{-1}$   
GJ-E  
 $A^{-1} \cdot A$   
 $A^{-1} \cdot b$   
 $A^{-1} \cdot I$

# Algorithm 2: 3 by 3 Example

**Problem:** Find the inverse of  $A = \begin{bmatrix} 1 & 6 & 0 \\ 1 & 7 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ .

**Solution:**

$$\begin{aligned}
 & \begin{array}{c} 7-6 \quad 1-0 \quad 0-1 \quad 1-0 \quad 0-0 \\ \left[ \begin{array}{ccc|ccc} \textcircled{1} & 6 & 0 & 1 & 0 & 0 \\ 1 & 7 & 1 & 0 & 1 & 0 \\ 0 & 1 & 3 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_2 \rightarrow (-1)R_1 + R_2} \left[ \begin{array}{ccc|ccc} 1 & 6 & 0 & 1 & 0 & 0 \\ 0 & \textcircled{1} & 1 & -1 & 1 & 0 \\ 0 & 1 & 3 & 0 & 0 & 1 \end{array} \right] \\
 & \quad \quad \quad \begin{array}{c} 2-0 \quad 1-0 \quad 3-1 \quad 0-(-1) \quad 0-1 \quad 1-0 \end{array}
 \end{array}
 \\
 & \xrightarrow{R_3 \rightarrow (-1)R_2 + R_3} \left[ \begin{array}{ccc|ccc} 1 & 6 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 2 & 1 & -1 & 1 \end{array} \right] \xrightarrow{R_3 \rightarrow (1/2)R_3} \left[ \begin{array}{ccc|ccc} 1 & 6 & 0 & 1 & 0 & 0 \\ 0 & 1 & \textcircled{1} & -1 & 1 & 0 \\ 0 & 0 & 1 & 1/2 & -1/2 & 1/2 \end{array} \right]
 \\
 & \xrightarrow{R_2 \rightarrow (-1)R_3 + R_2} \left[ \begin{array}{ccc|ccc} 1 & 6 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -3/2 & 3/2 & -1/2 \\ 0 & 0 & 1 & 1/2 & -1/2 & 1/2 \end{array} \right] \xrightarrow{R_1 \rightarrow (-6)R_2 + R_1} \left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & 10 & -9 & 3 \\ 0 & 1 & 0 & -3/2 & 3/2 & -1/2 \\ 0 & 0 & 1 & 1/2 & -1/2 & 1/2 \end{array} \right]
 \\
 & \quad \quad \quad \begin{array}{c} I_3 \quad A^{-1} \end{array}
 \end{aligned}$$

Conclusion: The inverse of A is  $A^{-1} = \begin{bmatrix} 10 & -9 & 3 \\ -3/2 & 3/2 & -1/2 \\ 1/2 & -1/2 & 1/2 \end{bmatrix}$ .

## Verify: Inverse is Correct [Optional]

**Problem:** Find the inverse of  $A = \begin{bmatrix} 1 & 6 & 0 \\ 1 & 7 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ .

Conclusion: The inverse of A is  $A^{-1} = \begin{bmatrix} 10 & -9 & 3 \\ -3/2 & 3/2 & -1/2 \\ 1/2 & -1/2 & 1/2 \end{bmatrix}$ .

If you are not sure about the solution, can verify:

**Verify [optional]:**

$$A = \begin{bmatrix} 1 & 6 & 0 \\ 1 & 7 & 1 \\ 0 & 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 10 & -9 & 3 \\ -3/2 & 3/2 & -1/2 \\ 1/2 & -1/2 & 1/2 \end{bmatrix}.$$

$$AB = \begin{bmatrix} 1 \cdot 10 + 6(-3/2) + 0(1/2) & 1(-9) + 6(3/2) + 0(-1/2) & 1 \cdot 3 + 6(-1/2) + 0(1/2) \\ 1 \cdot 10 + 7(-3/2) + 1(1/2) & 1(-9) + 7(3/2) + 1(-1/2) & 1 \cdot 3 + 7(-1/2) + 1(1/2) \\ 0 \cdot 10 + 1(-3/2) + 3(1/2) & 0(-9) + 1(3/2) + 3(-1/2) & 0 \cdot 3 + 1(-1/2) + 3(1/2) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3.$$

Thus the computed matrix is correct.

$$LU = A$$

# Judgement Questions

Q1 The expression of the inverse of  $A$  involves the inverse of certain elementary matrices.

$$A^{-1} = E_7 \bar{E}_{7-1} \dots \bar{E}_1$$

F

Q2 The expression of LU decomposition involves the inverse of certain elementary matrices.

L & U

T

$$L = E_1^{-1} \bar{E}_2^{-1} \dots \bar{E}_k^{-1}$$

Q3 When using Algorithm 1 to compute the inverse, you ~~do not need to~~ write down elementary matrices.

F

Q4 When using Algorithm 2 to compute the inverse, you do not need to write down elementary matrices.

T

(5/5, ✓)

# Summary of II. 1 Algorithms for Computing Inverse

1) **Algorithm 1**: compute  $A^{-1}$  by  $A^{-1} = E_p \dots E_2 E_1$ , if  $n$  pivots.

Here  $E_1, \dots, E_k, E_{k+1}, E_{k+2}, \dots, E_p$  are elementary matrices during GE  
(to get an identity matrix)

**Bottom line:**

Do you know how to get  $E_i, \forall i$ , and multiply matrices?

If so, then you know how to compute  $A^{-1}$

# Summary of II. 1 Algorithms for Computing Inverse

1) **Algorithm 1:** compute  $A^{-1}$  by  $A^{-1} = E_p \dots E_2 E_1$ .

Here  $E_1, \dots, E_k, E_{k+1}, E_{k+2}, \dots, E_p$  are elementary matrices during GE (to get an identity matrix)

**Bottom line:**

Do you know how to get  $E_i, \forall i$ , and multiply matrices?

If so, then you know how to compute  $A^{-1}$

2) **Algorithm 2:** compute  $A^{-1} = E_p \dots E_2 E_1 I_n$  by applying elementary operations to  $I_n$ .

Reminder:  
Inverse may not exist

**Bottom line:**

( $E_i$  not needed.)

Do you know how to conduct GJ-E (on many vectors together)?

If so, then you know how to compute  $A^{-1}$

[Remark: To understand why this method of  $[A|I] \rightarrow [I|A^{-1}]$  works, requires knowledge of elementary matrices and matrix representation of row operations.

But to compute the result, only need to know GJ-E.]

# Main Theorem of Lec 10

## Theorem 10.2 (Equivalent Conditions for Invertibility)

Let  $A \in \mathbb{R}^{n \times n}$ .

The following statements are equivalent:

1.  $A$  is invertible

2. The linear system  $A\mathbf{x} = \mathbf{0}$  has a unique solution  $\mathbf{x} = \mathbf{0}$

3.  $A$  is a product of elementary matrices

4. After GJE of  $A$ , the final form is  $I_n$

5. More ...

will be very  
useful in the  
coming weeks

We have proved the equivalence of 1,2,3,4.

# Concluding Section

# Summary Today

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Write your summary below.

One sentence summary:

Hw 2. sum

Detailed summary:

due date;

After vacation

# Summary Today (of Instructor)

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## One sentence summary:

We study the test conditions and computation of inverse.

## Detailed summary:

### 1. Test conditions

- **Algorithm test**:  $n$  pivots
- **Equation test**:  $Ax = 0$  has a unique solution
- Inverse can be written as product of elementary matrices

### 2. Expressions and computation of inverse

- Expression  $A^{-1} = E_p \dots E_2 E_1$ . (10.1)
- **Algorithm 1**: Use (10.1).
- **Algorithm 2**: apply GE to  $[A, I]$  to get  $[I, A^{-1}]$

### 3. Time complexity

- Vector addition and multiplication:  $O(n)$
- Matrix-vector multiplication:  $O(n^2)$
- Matrix-matrix multiplication:  $O(n^3)$
- Gaussian elimination:  $O(n^3)$

# Appendix: Comment on Proof of “Inverse Exists iff $n$ pivots”

Just for reading.

## Remark on Difficulties [extended reading]

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### First, right inverse.

Use GJ-E matrix representation, can only prove:

If there are  $n$  pivots, then there exists **left inverse** of  $A$ .

Need to: a) use GJE to  $[A, I]$  to show right inverse exists;

b) Then show left inverse = right inverse.

**Second**, not easy to prove the **reverse direction**:

If  $A$  is invertible, then there must be  $n$  pivots.

**Method 1 (Textbook):** prove by contradiction;  
requires 4 steps; requires deeper understanding of GE.

**Method 2 (next):** use PLU decomposition



# Summary Today (write Your Own)

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**One sentence summary:**

**Detailed summary:**